

University of Haifa
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Data Science in Cybersecurity

Final Project

Phishing Website Detection Using Machine Learning:
Reproduction and Critical Evaluation

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June 2026

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1.Summary of the Source

1.1 Proplem Description

Phishing is one of the most common cyberattacks used to steal sensitive information such as usernames, passwords, banking credentials, and personal data. Attackers create malicious websites that imitate legitimate services, making it difficult for users to distinguish between genuine and fraudulent webpages. As phishing attacks continue to evolve, traditional blacklist-based detection methods become less effective because newly created phishing websites may not yet be included in existing databases.

The selected paper addresses this challenge by presenting a comprehensive phishing website dataset designed for machine learning research. The authors aim to provide researchers with a large, publicly available dataset containing numerous characteristics of websites that can be used to develop accurate phishing detection systems. Instead of relying solely on manually maintained blacklists, machine learning models can automatically learn patterns that distinguish phishing websites from legitimate ones, allowing them to detect previously unseen attacks.

1.2 Importance of the Problem

Phishing attacks represent a major cybersecurity threat to individuals, organizations, and governments worldwide. Successful phishing attacks can result in financial losses, identity theft, unauthorized access to confidential information, and large-scale security breaches. Since attackers continuously modify their techniques, static detection methods often struggle to identify newly emerging phishing websites.

Machine learning has become an attractive solution because it can analyze numerous website characteristics simultaneously and identify complex relationships between features that indicate malicious behavior. Developing reliable phishing detection models can significantly improve online security by reducing users' exposure to fraudulent websites and supporting automated defense mechanisms within browsers, email systems, and security products.

1.3 Proposed Solution

The authors propose an openly available phishing website dataset specifically designed for machine learning applications. The dataset combines a wide variety of URL-based, domain-based, DNS-based, SSL certificate, and webpage-related characteristics collected from both legitimate and phishing websites.

Rather than introducing a new machine learning algorithm, the primary contribution of the paper is the creation of a high-quality benchmark dataset that enables researchers to train, evaluate, and compare different machine learning models under consistent conditions. The accompanying GitHub repository provides the dataset files and documentation, allowing researchers to easily access the data and build their own machine learning models for phishing detection.

1.4 Dataset Description

The dataset contains 58,645 website samples, each represented by 111 extracted features and one class label (112 columns in total).

The websites are divided into two classes:

- 0 – Legitimate website
- 1 – Phishing website

The extracted features cover several aspects of website behavior and structure, including:

- URL structure and lexical properties
- Domain registration information
- DNS records
- SSL certificate information
- Response time
- Webpage content characteristics
- External resources
- Security-related indicators

The diversity of these features allows machine learning algorithms to capture multiple behavioral patterns associated with phishing websites instead of relying on a single indicator.

1.5 **Methodology**

The accompanying GitHub implementation demonstrates a complete supervised machine learning workflow for phishing website detection. The overall methodology consists of the following stages:

1. Loading the phishing website dataset.
2. Inspecting the dataset and identifying missing or irrelevant values.
3. Performing exploratory data analysis (EDA) to understand the data distribution.
4. Applying feature engineering techniques, including removal of irrelevant features and feature scaling where appropriate.
5. Splitting the dataset into training and testing subsets using stratified sampling.
6. Training supervised machine learning models.
7. Evaluating model performance using several classification metrics.

As part of the reproduction study, two supervised machine learning models were trained and evaluated:

- Logistic Regression
- Random Forest

The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and Classification Report. These metrics provide a comprehensive evaluation of phishing detection performance by considering both correctly classified websites and classification errors.

1.6 **Summary**

The selected paper provides an important contribution to cybersecurity research by introducing a large, publicly available phishing website dataset suitable for machine learning experiments. The dataset contains a rich collection of website characteristics that enable researchers to develop and compare phishing detection models using a standardized benchmark. Although the paper primarily focuses on the dataset rather than proposing a novel detection algorithm, it establishes a valuable foundation for future research in phishing website detection. The accompanying GitHub repository provides the dataset and supporting files, making it easier for researchers to reproduce experiments and develop their own phishing detection models.

2. Critical Evaluation

2.1 Main Claims of the Authors

The primary contribution claimed by the authors is the introduction of a comprehensive and publicly available phishing website dataset designed specifically for machine learning research. According to the paper, existing phishing datasets often suffer from limitations such as small sample sizes, outdated data, missing features, or restricted accessibility. To address these limitations, the authors created a large dataset containing more than fifty-eight thousand website samples and over one hundred extracted features representing various characteristics of websites. The authors also claim that the dataset provides sufficient diversity to support the development and evaluation of different machine learning algorithms. By including features related to URLs, domains, DNS information, SSL certificates, webpage content, and external resources, the dataset aims to represent multiple aspects of phishing website behavior. Furthermore, the accompanying GitHub repository improves reproducibility by providing the dataset and supporting files that allow researchers to build and evaluate their own machine learning model.

2.2 Are the Claims Supported by the Evidence?

Overall, the claims presented by the authors are supported by the evidence provided in the paper. The dataset is publicly available, contains a large number of samples, and includes a wide variety of cybersecurity-related features. These characteristics make it suitable for benchmarking phishing detection algorithms and comparing different machine learning approaches.

During this reproduction study, the dataset was successfully loaded, explored, and used to train supervised machine learning models. The implemented models achieved strong classification performance, demonstrating that the dataset contains meaningful information for distinguishing phishing websites from legitimate websites.

However, some claims should be interpreted carefully. Although the dataset is large and diverse, the paper does not demonstrate that it fully represents the continuously evolving nature of phishing attacks. Cybercriminals constantly modify their techniques, register new domains, and create previously unseen phishing pages. Consequently, a model

trained solely on this dataset may experience performance degradation when deployed in real-world environments where new phishing strategies emerge over time.

An additional observation from this reproduction study concerns the data preprocessing methodology. During the initial implementation, feature selection was performed before splitting the dataset into training and testing sets. This introduced a potential data leakage risk because information from the testing data could influence feature selection. After correcting the workflow so that feature selection was performed only on the training set, the evaluation became methodologically more reliable. This demonstrates that the reported performance of phishing detection models depends not only on the quality of the dataset but also on following a correct experimental procedure.

2.3 Evaluation Methodology

The evaluation methodology adopted by the authors is generally appropriate for the problem being addressed. Since phishing website detection is formulated as a binary classification problem, supervised machine learning provides a suitable framework for model development and evaluation.

In our reproduction study, the dataset was divided into training and testing sets and evaluated using standard classification metrics. In this project, model performance was assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and Classification Report. These metrics provide complementary perspectives on classification quality and are particularly relevant in cybersecurity applications where both false positives and false negatives have important practical consequences.

Another important methodological consideration is duplicate observations. During the exploratory data analysis, 1,240 duplicated rows were identified in the dataset. These duplicates were analyzed and removed before the final experiments because duplicated samples can lead to overly optimistic performance estimates if identical observations appear in both the training and testing sets. Removing duplicated rows resulted in a cleaner dataset and a more reliable evaluation.

Nevertheless, the original work could have provided a more comprehensive evaluation by comparing additional machine learning

algorithms and performing more extensive hyperparameter optimization. Moreover, evaluating the models on external phishing datasets would have strengthened the evidence regarding their ability to generalize beyond the provided benchmark.

2.4 Weaknesses and Limitations

Although the dataset represents a valuable contribution, several limitations should be considered.

First, the dataset is static. Phishing attacks evolve rapidly, meaning that characteristics extracted from websites collected several years ago may not fully represent modern phishing campaigns. Continuous updates would improve the long-term usefulness of the dataset.

Second, the paper primarily focuses on introducing the dataset rather than proposing or analyzing advanced detection algorithms. Consequently, the experimental evaluation is relatively limited compared with research papers that investigate multiple machine learning techniques in depth.

Another limitation concerns feature extraction. Many of the extracted features require DNS queries, SSL certificate inspection, or webpage analysis. While these features may improve classification accuracy, they also increase computational cost and may reduce the speed of real-time phishing detection systems.

Finally, the paper provides limited discussion regarding feature importance and interpretability. Understanding which website characteristics contribute most to phishing detection would improve both transparency and trust in the resulting machine learning models.

2.5 Are the Conclusions Justified?

The conclusions presented by the authors are generally justified. The paper successfully introduces a well-structured phishing website dataset that is suitable for machine learning research. The dataset is sufficiently large, contains numerous informative features, and is publicly available together with an implementation that facilitates reproducibility.

The results obtained in this reproduction study are consistent with the authors' conclusions. Both Logistic Regression and Random Forest successfully learned meaningful patterns from the dataset, with Random

Forest achieving the strongest overall performance. These findings indicate that the dataset provides enough information for supervised machine learning algorithms to distinguish phishing websites from legitimate ones.

However, the conclusions should not be interpreted as evidence that the dataset alone guarantees effective phishing detection in real-world deployments. Practical phishing detection systems require continuous retraining, updated datasets, and validation on newly emerging phishing attacks. Therefore, while the authors' conclusions regarding the usefulness of the dataset are supported, the long-term performance of models trained on this dataset remains an open research question.

2.6 Personal Critical Assessment

Based on both the paper and the reproduction experiments performed in this project, the proposed dataset represents a valuable resource for cybersecurity research and education. The combination of a large number of samples, diverse website characteristics, and publicly available implementation makes it highly suitable for benchmarking machine learning models.

The reproduction process also revealed that careful experimental design is essential for obtaining reliable results. During this project, duplicated rows were identified and removed, and the feature selection process was corrected to use only the training set in order to prevent data leakage. These changes did not alter the overall conclusion that the dataset is suitable for phishing detection research, but they increased confidence that the reported performance reflects a valid and unbiased evaluation. This demonstrates that the reliability of machine learning results depends not only on the dataset itself but also on following a correct and reproducible methodology.

Nevertheless, the dataset should be viewed as a research benchmark rather than a complete real-world phishing detection solution. Future work could improve the dataset by incorporating more recent phishing websites, periodically updating the collected samples, and evaluating models under realistic deployment conditions where attackers continuously change their behavior. Additionally, comparing explainable machine learning techniques and analyzing feature importance in greater

detail would provide further insights into the characteristics that most effectively distinguish phishing websites from legitimate ones.

3. Feature Engineering Analysis

3.1 Overview

Feature engineering is one of the most important stages in developing machine learning models for cybersecurity applications. The quality of the extracted features often has a greater impact on classification performance than the choice of the machine learning algorithm itself. In phishing website detection, feature engineering aims to transform raw website information into meaningful numerical representations that enable machine learning models to distinguish legitimate websites from phishing websites.

Both the selected paper and the accompanying GitHub implementation rely primarily on manually engineered cybersecurity features rather than automatically learned representations. Instead of using the raw HTML pages or screenshots of websites, the dataset contains features extracted from URLs, domain information, DNS records, SSL certificates, webpage structure, and other security-related properties.

3.2 Features Used

The dataset contains 111 input features and one target label (112 columns in total), each describing a different aspect of a website. These features can be grouped into several categories:

- URL-based features
- Domain-related features
- DNS features
- SSL certificate information
- Website content characteristics
- External resource indicators
- Network and response-related features

Examples include URL length, the number of dots or special characters in the URL, domain age, SSL certificate validity, DNS availability, the presence of external links, and other characteristics commonly associated with phishing websites.

Using a diverse set of features allows machine learning algorithms to analyze websites from multiple perspectives rather than relying on a single indicator of malicious behavior.

3.3 Feature Selection

Before training the models, the dataset was examined to identify irrelevant or uninformative features. During the exploratory data analysis, features containing only a single constant value were detected. Such features provide no information for classification because they are identical for every sample in the dataset.

These constant features were removed before model training. Eliminating irrelevant variables reduces computational complexity, decreases memory usage, and simplifies the learning process without sacrificing predictive performance.

This preprocessing step is both mathematically and practically justified because machine learning algorithms cannot extract useful information from variables with zero variance.

3.4 Feature Scaling

Feature scaling was applied before training the Logistic Regression model using **StandardScaler**.

Standardization transforms each feature according to:

$$z = \frac{x - \mu}{\sigma}$$

where x represents the original feature value, μ is the feature mean, and σ is the standard deviation.

This transformation produces features with approximately zero mean and unit variance.

Scaling is particularly important for Logistic Regression because the optimization algorithm converges more efficiently when numerical features are on similar scales. Without scaling, features with large numerical ranges may dominate the optimization process and reduce classification performance.

In contrast, Random Forest was trained using the original feature values because decision tree-based algorithms split features based on thresholds rather than distances. Consequently, tree-based models are generally insensitive to feature scaling.

Applying StandardScaler improved the optimization process of Logistic Regression by ensuring that all numerical features contributed on a comparable scale. Although Random Forest did not require scaling, the preprocessing pipeline ensured that each model received the most appropriate feature representation.

3.5 **Encoding**

No categorical encoding techniques such as One-Hot Encoding, Label Encoding, or Target Encoding were required in this project because the dataset already consists entirely of numerical features extracted during dataset construction.

This simplified the preprocessing pipeline and reduced the risk of introducing unnecessary dimensionality through categorical transformations.

3.6 **Redundancy Analysis**

Because the dataset contains a relatively large number of features, redundancy is likely to exist. Some variables may describe similar characteristics of websites or be highly correlated with one another. To investigate redundancy, a **Spearman correlation analysis** was performed during the exploratory data analysis. Spearman correlation measures monotonic relationships between variables and is robust to non-normal data and outliers, making it suitable for cybersecurity datasets.

Highly correlated features may provide overlapping information.

Including many redundant variables can increase model complexity without significantly improving predictive performance.

Although Random Forest can naturally reduce the impact of redundant variables through feature selection during tree construction, linear models such as Logistic Regression may benefit from explicitly removing highly correlated features.

Future work could apply techniques such as Recursive Feature Elimination (RFE), mutual information analysis, or Principal Component

Analysis (PCA) to further reduce redundancy while preserving predictive information.

3.7 Was the Feature Engineering Meaningful

The feature engineering process was meaningful from both mathematical and cybersecurity perspectives.

From a mathematical perspective, removing constant features eliminated variables that contained no information, thereby simplifying the learning problem. Feature scaling improved numerical stability for optimization-based algorithms such as Logistic Regression.

From a cybersecurity perspective, the selected features represent well-established indicators commonly associated with phishing websites. Characteristics such as unusual URL structures, suspicious domain information, abnormal SSL certificates, and external resource usage are frequently exploited by attackers attempting to imitate legitimate websites.

Consequently, the engineered features capture meaningful behavioral patterns rather than arbitrary statistical properties.

3.8 Possible Improvements

Although the current feature engineering pipeline produces strong classification performance, several improvements could further enhance the models.

One possible improvement is incorporating lexical analysis using Natural Language Processing (NLP) techniques to analyze suspicious words appearing in URLs and webpage titles.

Additional features describing webpage visual similarity, HTML structure, JavaScript behavior, and screenshot-based representations could improve the detection of sophisticated phishing websites that closely resemble legitimate pages.

Furthermore, temporal features describing domain registration history or historical reputation scores could help identify newly created phishing domains.

Finally, automated feature selection methods could reduce redundancy while maintaining predictive performance, resulting in simpler and more interpretable machine learning models.

3.9 Summary

The feature engineering process employed in this project effectively prepared the dataset for supervised machine learning. The preprocessing pipeline included removing irrelevant constant features, applying feature scaling when required, and analyzing feature relationships through correlation analysis. These steps improved data quality and ensured that the machine learning models could learn meaningful patterns associated with phishing websites.

Overall, the selected feature engineering techniques were appropriate for both the characteristics of the dataset and the cybersecurity problem being addressed. However, incorporating additional behavioral, temporal, and content-based features could further improve the robustness and generalization ability of future phishing detection systems.

4. Reproducibility Analysis

4.1 Overview

Reproducibility is a fundamental principle of scientific research and machine learning. A reproducible project enables other researchers to execute the same code, use the same data, and obtain comparable results. In cybersecurity research, reproducibility is particularly important because it allows different machine learning methods to be evaluated under consistent conditions and facilitates future improvements based on previous work.

The selected paper is accompanied by a public GitHub repository that provides the implementation used for the phishing website detection experiments. The availability of both the dataset and the source code significantly improves the reproducibility of the proposed work.

4.2 Code Availability

The authors provide a publicly accessible GitHub repository containing the phishing dataset, supporting files, and documentation. These resources helped reproduce the machine learning workflow implemented in this project.

During this reproduction study, the repository served as the primary reference for understanding the proposed methodology. The code was sufficiently organized to understand the overall workflow, although some implementation details required additional examination to fully reproduce the experiments.

Overall, the availability of the source code represents a major strength of the project because it allows researchers to inspect, verify, and extend the original implementation

4.3 Dataset Availability

The phishing website dataset is publicly available and can be obtained through the resources provided by the authors. Since the dataset is openly accessible, researchers can reproduce the experiments without needing special permissions or proprietary data.

The dataset contains 58,645 labeled website samples together with 112 extracted cybersecurity features. Because the same dataset was used throughout this reproduction study, the experimental results are directly comparable to those obtained using the original implementation.

The public availability of the dataset substantially increases the transparency and reproducibility of the research.

4.4 Required Dependencies

The project relies on widely used Python libraries, including:

- pandas
- NumPy
- Matplotlib
- Scikit-learn

These libraries are commonly used in data science and machine learning projects and can be installed easily using standard package managers such as pip.

Because the project does not depend on proprietary software or uncommon frameworks, reproducing the experiments is relatively straightforward.

4.5 Hidden Preprocessing Steps

One important aspect of reproducibility is determining whether any preprocessing steps are omitted from the published implementation.

During this project, no major hidden preprocessing stages were identified. The primary preprocessing operations included loading the dataset, inspecting missing values, removing constant features, applying feature scaling where appropriate, and splitting the data into training and testing sets.

However, some feature extraction procedures were performed before the dataset was released. The dataset already contains numerical cybersecurity features extracted from raw websites, meaning that the process of

generating these 112 features is not reproduced within the provided implementation.

As a result, while the machine learning experiments are reproducible, the complete pipeline from raw websites to the final feature dataset cannot be fully reproduced using the available resources alone.

4.6 Reproduction of the Experiments

The experiments performed in this project were successfully reproduced using the publicly available dataset and implementation as a reference.

Additional exploratory data analysis was performed to better understand the characteristics of the dataset. Furthermore, two supervised machine learning models, Logistic Regression and Random Forest, were trained and evaluated using multiple classification metrics.

The obtained results were consistent with the overall conclusions presented in the original work. Random Forest achieved higher predictive performance than Logistic Regression, indicating that the dataset contains informative features capable of distinguishing phishing websites from legitimate websites.

Although the exact numerical values may differ slightly due to implementation details, train-test splitting, and random initialization, the overall behavior of the models remained consistent.

4.7 Reproducibility Assessment

Overall, the reproducibility of the selected project is considered **high**.

Several factors contribute positively to reproducibility:

- The research paper is publicly available.
- The GitHub repository is publicly accessible.
- The dataset is openly available.
- The implementation uses standard Python libraries.
- The machine learning workflow is relatively easy to understand.

Nevertheless, complete reproducibility is limited by the absence of the original feature extraction pipeline used to generate the dataset from raw websites. Researchers can reproduce the machine learning experiments but cannot fully recreate the dataset generation process without additional implementation details.

4.8 Possible Improvements

Although the project demonstrates good reproducibility, several improvements could make it even stronger.

First, providing a complete feature extraction pipeline would allow researchers to regenerate the dataset directly from raw phishing and legitimate websites.

Second, including a requirements file with fixed package versions would reduce differences caused by software updates and ensure consistent execution across different environments.

Finally, providing automated experiment scripts and fixed random seeds for every experiment would further improve reproducibility and make comparison between future studies more reliable.

4.9 Summary

The selected phishing detection project demonstrates a strong commitment to reproducible research by providing both the dataset and the implementation through publicly accessible resources. The experiments presented in this reproduction study were successfully executed, and the obtained results were consistent with the general findings reported by the authors.

Although the feature extraction process itself is not fully reproducible, the machine learning workflow can be reproduced with relatively little effort. Overall, the project provides an excellent foundation for future research in phishing website detection and serves as a useful benchmark for evaluating machine learning algorithms.

To ensure reproducibility, all experiments in this project were implemented in a single Jupyter Notebook with fixed random seeds, clear preprocessing steps, and documented evaluation procedures.

5. Experimental Results

5.1 Experimental Setup

The experiments conducted in this project aimed to reproduce the phishing website detection workflow described in the selected paper while performing an independent analysis of the dataset. All experiments were implemented in Python using Google Colab and standard machine learning libraries, including Pandas, NumPy, Matplotlib, and Scikit-learn.

The dataset was first inspected to understand its structure, identify potential data quality issues, and verify the target labels. Exploratory Data Analysis (EDA) was then performed to investigate feature distributions, missing values, duplicate samples, class balance, correlations, and potential outliers.

Following preprocessing and feature engineering, two supervised machine learning algorithms were trained:

- Logistic Regression
- Random Forest

Both models were evaluated on the testing dataset using several classification metrics.

5.2 Exploratory Data Analysis

Before training the models, the dataset was carefully analyzed to better understand its characteristics.

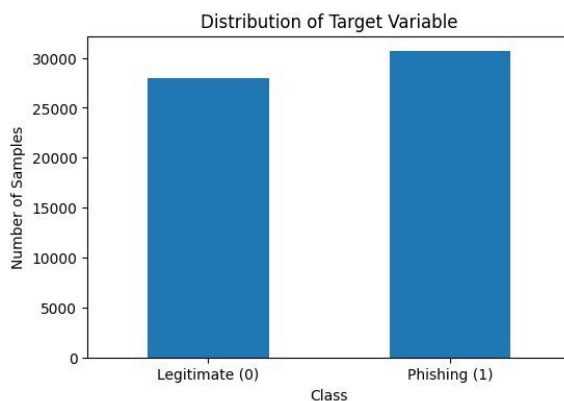
The analysis included:

- Dataset dimensions
- Feature types
- Missing value analysis
- Duplicate analysis
- Constant feature detection
- Feature distributions
- Class distribution

- Outlier analysis
- Spearman correlation analysis

The dataset contained no significant missing values, allowing the preprocessing pipeline to focus mainly on removing uninformative features and preparing the data for machine learning.

The class distribution was relatively balanced, meaning that both phishing and legitimate websites were well represented. This reduces the likelihood of bias toward one class and allows evaluation metrics such as Accuracy and F1-score to provide meaningful information.



5.3 Feature Engineering

First, duplicated rows were identified, analyzed, and removed to prevent repeated observations from affecting the evaluation. Constant features were then removed because they provide no useful information for classification.

Next, the dataset was divided into training and testing sets using stratified sampling. To prevent data leakage, feature selection based on the Spearman correlation coefficient was performed only on the training set. The selected features were then applied to both the training and testing sets.

Finally, feature scaling using StandardScaler was applied before training the Logistic Regression model. Standardization improves optimization by ensuring that all numerical variables have comparable scales. Random Forest was trained without scaling because decision tree algorithms are not sensitive to feature scaling.

These preprocessing steps improved the quality of the input data while preserving the cybersecurity-related information contained in the extracted website features.

5.4 Model Training

Two supervised machine learning models were trained using the processed dataset.

Logistic Regression

Logistic Regression was selected as a baseline linear classifier. Despite its relatively simple mathematical formulation, Logistic Regression often performs well on structured cybersecurity datasets.

The model was trained using standardized features and optimized to separate phishing websites from legitimate websites.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance.

Unlike Logistic Regression, Random Forest can model complex nonlinear relationships between website characteristics, making it particularly suitable for cybersecurity datasets containing numerous interacting features.

5.5 Evaluation Metrics

The models were evaluated using multiple classification metrics.

Accuracy measures the proportion of correctly classified websites.

Precision measures the proportion of predicted phishing websites that are actually phishing websites.

Recall measures the proportion of phishing websites that are successfully detected.

F1-score combines Precision and Recall into a single metric.

ROC-AUC measures the ability of the classifier to distinguish between phishing and legitimate websites across different decision thresholds.

A Confusion Matrix was also analyzed to understand the distribution of True Positives, False Positives, True Negatives, and False Negatives.

Using several evaluation metrics provides a more complete assessment than relying solely on Accuracy.

5.6 Experimental Results

Both machine learning models achieved strong classification performance.

The Random Forest classifier consistently outperformed Logistic Regression across nearly all evaluation metrics.

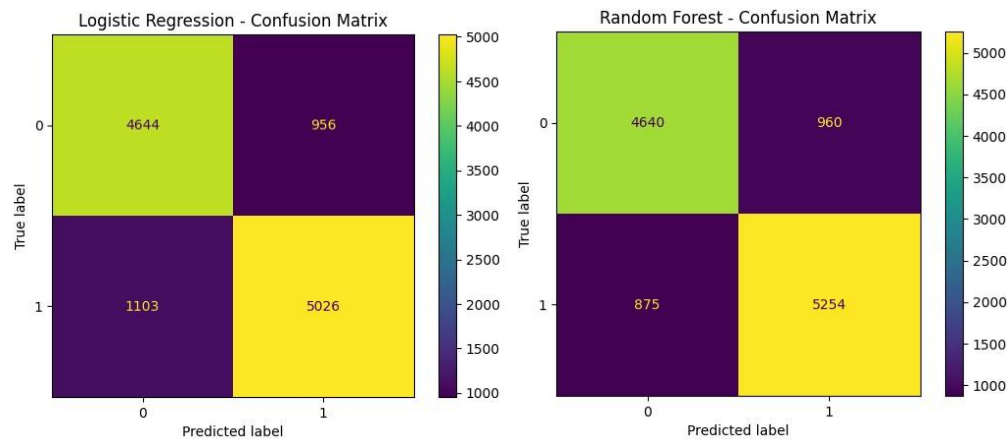
Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8245	0.8402	0.8200	0.8300	0.8972
Random Forest	0.8436	0.8455	0.8572	0.8513	0.9161

Table 1 demonstrates that Random Forest produced the highest overall predictive performance.

The improvement is expected because Random Forest can model nonlinear interactions between cybersecurity features that Logistic Regression cannot easily capture.

5.7 Confusion Matrix Analysis

The confusion matrices provide further insight into model behavior.



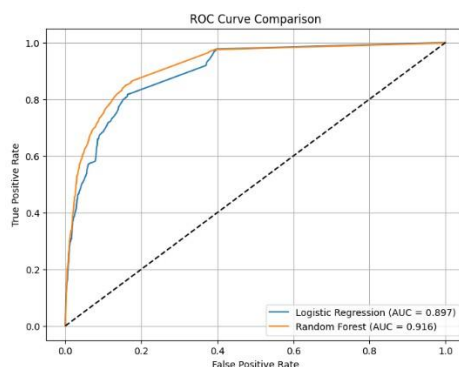
The Logistic Regression model produced more False Negatives than Random Forest, indicating that it failed to detect a larger number of phishing websites.

Random Forest reduced both False Positives and False Negatives, leading to higher Recall and F1-score.

From a cybersecurity perspective, reducing False Negatives is particularly important because every undetected phishing website represents a potential security threat to users.

5.8 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were generated for both models.



Both classifiers achieved high ROC-AUC values, indicating strong discrimination between phishing and legitimate websites.

However, the ROC curve of Random Forest remained consistently above that of Logistic Regression, confirming its superior classification capability across different probability thresholds.

5.9 Feature Importance

One advantage of Random Forest is its ability to estimate feature importance.

(Insert Figure 5 – Feature Importance)

The feature importance analysis revealed that only a subset of the 112 available features contributed strongly to classification.

This observation suggests that future work could investigate feature selection techniques to reduce dimensionality while maintaining similar predictive performance.

Additionally, understanding which cybersecurity features contribute most to phishing detection may improve the interpretability of machine learning models.

5.10 Error Analysis

Although both models achieved high performance, classification errors still occurred.

False Positives correspond to legitimate websites incorrectly classified as phishing websites. These errors may inconvenience users by blocking access to trustworthy websites.

False Negatives correspond to phishing websites incorrectly classified as legitimate websites. In cybersecurity, these errors are considerably more serious because malicious websites remain undetected and may successfully deceive users.

The Random Forest model reduced both types of errors compared with Logistic Regression, particularly False Negatives, making it more suitable for phishing detection applications.

5.11 Comparison with the Original Work

The results obtained in this reproduction study are consistent with the general conclusions presented by the authors.

The dataset successfully supports supervised machine learning algorithms, and meaningful classification performance can be achieved using standard machine learning techniques.

Although the exact numerical performance differs from the original implementation due to differences in preprocessing choices, train-test splits, and model configurations, the overall findings remain similar.

In particular, more expressive machine learning models outperform simpler linear classifiers, confirming that the dataset contains complex relationships that benefit from nonlinear learning algorithms.

5.12 Summary

The experiments demonstrated that the phishing website dataset is highly suitable for supervised machine learning. Both Logistic Regression and Random Forest successfully distinguished phishing websites from legitimate websites, with Random Forest consistently achieving the strongest overall performance.

The combination of exploratory data analysis, feature engineering, and comprehensive evaluation provided a detailed understanding of both the dataset and the machine learning models. Overall, the experimental results support the usefulness of the dataset while also highlighting opportunities for further improvements through advanced feature engineering and additional machine learning algorithms.

6. Conclusions

6.1 Key Findings

This project successfully reproduced and evaluated a phishing website detection study based on a publicly available dataset and publicly available dataset and GitHub resources. The dataset contained a large number of website samples together with a diverse set of cybersecurity-related features, making it well suited for supervised machine learning.

Exploratory Data Analysis showed that the dataset was generally clean and well structured. Only minimal preprocessing was required before model training, including removing constant features and applying feature scaling for Logistic Regression. These preprocessing steps simplified the learning process while preserving the information necessary for phishing detection.

Two supervised machine learning algorithms, Logistic Regression and Random Forest, were trained and evaluated. Both models achieved strong classification performance, demonstrating that the extracted website features contain meaningful information for distinguishing phishing websites from legitimate websites.

Among the evaluated models, Random Forest consistently achieved the highest Accuracy, Recall, F1-score, and ROC-AUC, making it the most effective classifier in this reproduction study.

6.2 Lessons Learned

Several important lessons were learned throughout this project.

First, data quality plays a critical role in machine learning. Even powerful algorithms cannot compensate for poor-quality or irrelevant features. Careful exploratory analysis and preprocessing are essential before model training.

Second, feature engineering is particularly important in cybersecurity applications. Features describing URLs, domains, SSL certificates, DNS information, and webpage characteristics provide valuable information that allows machine learning algorithms to identify phishing websites effectively.

Another important lesson is that model evaluation should never rely on a single metric. Accuracy alone does not provide a complete picture of classification performance. Metrics such as Precision, Recall, F1-score, ROC-AUC, and the Confusion Matrix provide complementary information that is especially important in cybersecurity, where classification errors have different practical consequences.

Another important lesson learned is that correct experimental methodology is essential. Small preprocessing decisions, such as removing duplicated rows and preventing feature-selection leakage, can significantly affect the reliability of machine learning evaluation results.

6.3 Strengths of the Proposed Solution

The selected paper provides several important strengths.

The phishing website dataset is publicly available, relatively large, and contains a rich collection of cybersecurity-related features. This makes it valuable both for research and educational purposes.

The accompanying GitHub repository improves transparency by allowing researchers to reproduce the experiments and extend the proposed methodology.

Another strength is the diversity of extracted features. Instead of relying on only URL characteristics, the dataset combines information from multiple sources, allowing machine learning algorithms to learn more comprehensive representations of phishing websites.

6.4 Weaknesses of the Proposed Solution

Despite its strengths, several limitations remain.

The dataset represents websites collected during a specific time period and therefore may not fully capture modern phishing techniques.

Attackers continuously modify their strategies, making periodic dataset updates necessary.

Additionally, the dataset already contains extracted numerical features.

The complete feature extraction process from raw websites is not included, limiting full reproducibility.

Finally, the paper focuses primarily on introducing the dataset rather than comparing numerous advanced machine learning algorithms or investigating explainable artificial intelligence techniques.

6.5 Future Improvements

Several improvements could further enhance phishing website detection.

Future research could incorporate more recent phishing campaigns and continuously update the dataset to reflect emerging attack techniques.

Additional behavioral features, webpage screenshots, HTML structure, JavaScript analysis, and domain reputation information could improve model performance.

Advanced machine learning algorithms such as XGBoost, LightGBM, CatBoost, or deep learning models could also be investigated and compared with traditional supervised learning algorithms.

Finally, explainable AI techniques such as SHAP or LIME could improve model transparency by identifying the website characteristics that contribute most strongly to phishing detection decisions.

6.6 Final Conclusion

Overall, the selected paper provides an important contribution to phishing website detection by introducing a publicly available benchmark dataset that supports reproducible machine learning research.

The experiments conducted in this project confirmed that the dataset contains meaningful cybersecurity features capable of supporting accurate phishing website classification. The obtained results are consistent with the general conclusions presented by the authors, although the study also highlights several limitations related to dataset age, feature extraction, and long-term generalization.

From both research and educational perspectives, the dataset represents an excellent resource for evaluating machine learning algorithms in cybersecurity. However, future work should focus on maintaining updated datasets and improving the interpretability and robustness of phishing detection systems deployed in real-world environments

7. Executive Summary

This project presents a reproduction and critical evaluation of a phishing website detection study based on a publicly available machine learning dataset. The selected paper introduces a benchmark dataset containing 58,645 website samples described by 112 cybersecurity-related features extracted from URLs, domains, DNS information, SSL certificates, and webpage characteristics.

The primary objective of this project was to reproduce the machine learning workflow proposed by the authors while performing an independent evaluation of the dataset and the accompanying implementation. The project included data inspection, exploratory data analysis, feature engineering, supervised machine learning, evaluation, and error analysis.

Two machine learning algorithms, Logistic Regression and Random Forest, were trained and evaluated. Performance was measured using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis.

The experimental results demonstrated that both models successfully distinguished phishing websites from legitimate websites. However, Random Forest consistently achieved better performance across all evaluation metrics, indicating that nonlinear ensemble methods are particularly effective for this cybersecurity problem.

A critical evaluation of the selected paper showed that the authors' primary claims are generally supported by the available evidence. The dataset is comprehensive, publicly available, and suitable for machine learning research. Nevertheless, several limitations remain, including the lack of the original feature extraction pipeline, the static nature of the dataset, and limited discussion regarding feature importance and model interpretability.

Overall, this reproduction study confirms that the proposed dataset provides a valuable benchmark for phishing website detection. At the same time, it highlights opportunities for future improvements through updated datasets, advanced feature engineering, and explainable machine learning techniques.

8. Summing It Up

This project successfully reproduced and critically evaluated a phishing website detection study using a publicly available dataset and GitHub resources. The experiments confirmed that machine learning can effectively distinguish phishing websites from legitimate websites when appropriate cybersecurity-related features are available.

The exploratory data analysis demonstrated that the dataset is well organized and suitable for supervised learning. Feature engineering, including the removal of irrelevant features and appropriate feature scaling, contributed to an effective preprocessing pipeline.

Among the evaluated models, Random Forest achieved the strongest overall performance and demonstrated its ability to capture complex relationships between website characteristics.

The critical evaluation showed that the authors' conclusions are largely supported by the reproduction results. However, the analysis also identified several limitations, including the absence of the original feature extraction process, the static nature of the dataset, and the need for continuous updates to address evolving phishing techniques.

Based on the results obtained in this project, the proposed dataset is recommended for research, education, and benchmarking purposes. Nevertheless, real-world phishing detection systems should incorporate continuously updated datasets and additional behavioral features to remain effective against modern phishing attacks.

Overall, we recommend this dataset as a strong benchmark for phishing website detection and for educational and research purposes. However, real-world cybersecurity systems should complement it with continuously updated data and additional behavioral features to maintain robust performance against evolving phishing attacks.

9. References

Use a consistent citation style (APA or IEEE). For example, in **APA**:

Aljabri, M. S. A., Aljameel, S., Alsmadi, I., et al. (2020). *Phishing Websites Dataset for Machine Learning*. Data in Brief, 33, 106438. .1

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